



High Performance Computing

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Outline

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- Course Details
- Syllabus
- Evaluation Plan
- Reference Materials
- SUPERCOMPUTERS

- What is computer Program

- Algorithm and data structure.
- Programming Language: Standard Notation for writing programs
 - Example : C, Java, Intel assembly language
 - **Machine Language(IA32).**
- Program translators : Compiler - GCC
 - What does GCC do?

- How Data Represented?

- Binary
 - Base2/0-1/bit

Improving of Computer Performance :

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- Innovative manufacturing and advancement of VLSI.
- Exploitation of some form parallelism
 - ILP
 - Pipelining (IE=F/D/E/M/)
 - Out of order execution
 - Superscalar
 - Thread Level Parallelism
 - Multiple thread on one processor
 - SMT- Hardware Multithreading.
 - Multithreading on Multiple processor.
 - Example : Loop
- Process Level Parallelism
 - Different processes can be executed in parallel on Multiple process
 - SMP
 - DSM

Syllabus

Module 1: Introduction:

Motivating Parallelism, Scope of Parallel Computing, Introduction to HPC: Parallel Programming Platforms; Implicit Parallelism: Trends in Microprocessor Architectures, Limitations of Memory System Performance, Dichotomy of Parallel Computing Platforms, Physical Organization of Parallel Platforms; .

Module 2: Parallel Computer Memory Architectures:

Measures of Parallel Algorithms; Analytical Modeling of Parallel Programs: Sources of Overhead in Parallel Programs, Performance Metrics for Parallel Systems, the Effect of Granularity on Performance; Parallel Platforms: Models (SIMD, MIMD, SPMD) , Communication (Shared Address Space vs. Message Passing)

Module 3 Threads :

Thread Basics: Why Threads? The POSIX Thread API, Thread Creation and Termination, Synchronization Primitives in Pthreads, Controlling Thread and Synchronization Attributes, Thread Cancellation, Composite Synchronization.

Module 4: OpenMP

Tips for Designing Asynchronous Programs, OpenMP: A Standard for Directive Based Parallel Programming.

Module 5 Distributed Memory Parallel Programming

The Building Blocks: Send and Receive Operations, MPI: the Message Passing Interface, Topologies and Embedding, Overlapping Communication with Computation, Collective Communication and Computation Operations, Groups and Communicators.

Module 5 GPGPU:

The Age of Parallel Processing, Central Processing Units, The Rise of GPU Computing, A brief history of GPUs, Early GPU computing; CUDA: What is CUDA architecture, using the CUDA architecture, Applications of CUDA, Medical Imaging, Computational Fluid Dynamics, Environmental Science; Introduction to CUDA C: A First Program, Hello world, A kernel call, Passing parameters, Querying devices, using device properties; Parallel Programming in CUDA C: CUDA parallel programming, Summing vectors, A fun example.

TEXT BOOKS

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- John Hennessy and David Patterson, Computer Architecture - A Quantitative Approach. 5ed. Morgan Kaufmann.
- Ananth Grama, Vipin Kumar, Anshul Gupta, George Karypis, Introduction to Parallel Computing, Addison-Wesley, 2003
- Wen-Mei W Hwu, David B Kirk, Programming Massively Parallel Processors A Hands-on Approach, Morgan Kaufmann, 3e.
- William Dally and Brian Towles. Principles and Practices of Interconnection Networks, MK, 2004

EVALUATION PLAN

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Mid-Evaluation	End Evaluation	Class/Tutorial-Participat	Quiz	Mini Project	Total
25	40	5	10	20	100

Week	Components
week 4	P-0/A-0
week 6	Quiz-1
Week-7	P1
week 8	Mid-Exam
week 15	Quiz-2
Week15	P1